IMDb Hybrid Dataset for LLM-Augmented Query Execution

Dataset Construction, Exploratory Analytics, and Fit to SUQL and Stretto

What Data Do SUQL and Stretto Expect?

SUQL: SQL + semantic text primitives

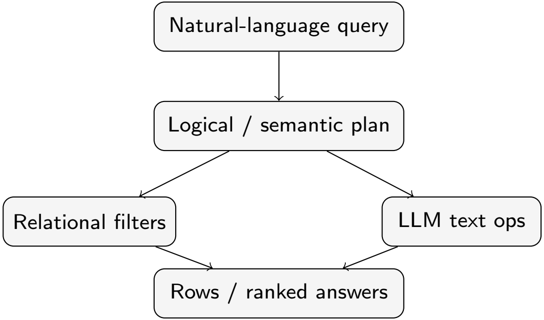
SUQL expects a relational schema where some columns are classical structured attributes and some columns are free-text fields.

Structured columns	Free-text columns
rating, year, runtime	reviews
genre, director, location	passages

The natural-language question is parsed into a SQL-like query using operators such as `ANSWER()` and `SUMMARY()`.

Stretto: LLM-augmented query execution

Stretto expects heterogeneous data sources that can be represented as structured relations plus semantic text evidence.



Dataset Construction Pipeline

Goal: create a compact hybrid benchmark from movie reviews and movie metadata.

Source 1: review corpus

- 25,000 IMDb-style user reviews
- free-form opinions and subjective descriptions
- text contains sentiment, plot comments, atmosphere, acting, pacing

Source 2: IMDb metadata

- title basics: title, year, runtime, genres
- title crew: director identifiers
- name basics: director names

Join key: `movie_id / tconst` Review rows + metadata tables → **joined hybrid table**

Original Datasets Before Joining

Initial inputs used in `data_preprocessing.ipynb`

1. Review dataset

Contains:

- IMDb review text
- movie identifier: `movie_id` or `tconst`

Purpose: provide the unstructured textual component.

2. IMDb metadata tables

File	Used columns
`title.basics.tsv`	title, year, runtime, genres
`title.crew.tsv`	movie-to-director ids
`name.basics.tsv`	director id-to-name mapping

Purpose: provide structured movie attributes.

Processing sketch

Normalize review columns → join with `title.basics` on `tconst` → filter `title.crew` to used movies → resolve director IDs through `name.basics` → export final joined CSV

Final Dataset Schema and Why It Fits

Column	Type
`movie_id`	identifier
`title`	text
`director`	categorical
`year`	numeric
`runtime`	numeric
`genres`	multi-label categorical
`review`	free text

Why this is a good fit

- one row combines movie metadata and review evidence
- structured predicates can reduce candidate sets before LLM calls
- review text supports subjective predicates such as *interesting*, *underrated*, *slow*, or *romantic*

Example hybrid query:

Find action movies released in 1998 whose reviews describe them as exciting.

Mapping the Dataset to SUQL and Stretto

SUQL-style execution

- natural-language question is parsed into SQL-like query
- `year`, `runtime`, `genres`, `director` are standard predicates
- `review` is queried with `ANSWER()` or `SUMMARY()`

```
SELECT title, summary(review)
FROM movies
WHERE year = 1998 AND 'Action' IN genres
AND answer(review, 'is it exciting?') = 'Yes';
```

Stretto-style execution

- logical plan mixes relational and semantic operators
- optimizer can choose cheaper semantic filters after structured pruning
- runtime–accuracy trade-offs can be evaluated over review-based predicates

```
Filter[year=1998, genre=Action] →
SemanticFilter[review: exciting] →
Rank/Limit[Top-3]
```

The dataset meets the core input requirements of both systems.

Dataset at a Glance

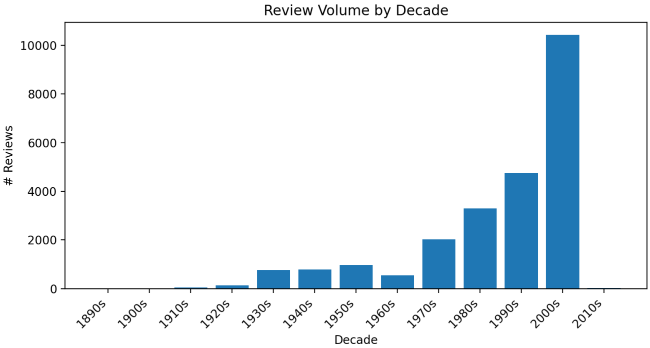
Metric	Value
Total reviews	25,000
Unique movies	3,456
Unique directors	2,690
Genre combinations	440
Year range	1895–2011
Median runtime	95 min
Median review length	173–174 words
Corpus size	∼5.8M words
Rows with HTML breaks	58.7%

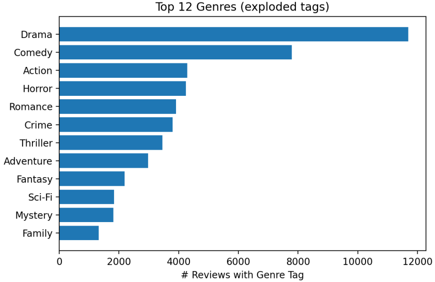
Data quality

Column	Missing
`title`	4.9%
`director`	5.1%
`year`	4.9%
`runtime`	5.7%
`genres`	4.9%
`review`	0.0%

The textual component is complete; metadata is mostly complete and usable for structured pruning.

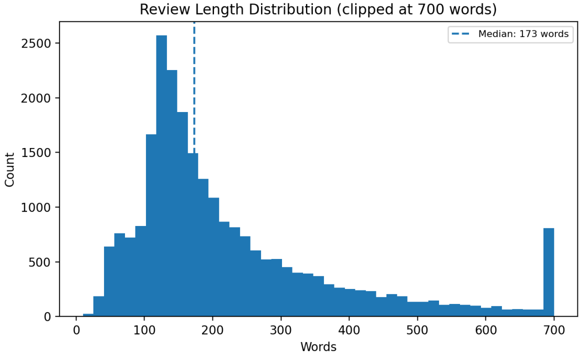
Balance Analysis: Time and Genre Skew

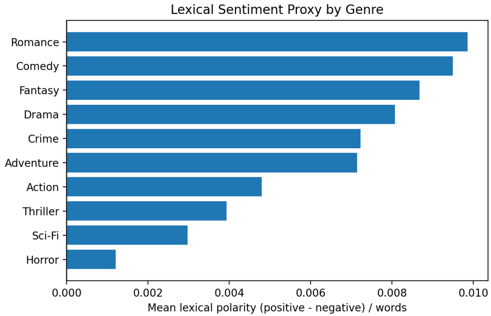




Interpretation. The dataset is clearly imbalanced. The 2000s dominate with 10,432 reviews, and Drama is far more frequent than most other genres. This is realistic for retrieval/recommendation settings, but evaluation should stratify by decade and genre to avoid overestimating performance on dominant slices.

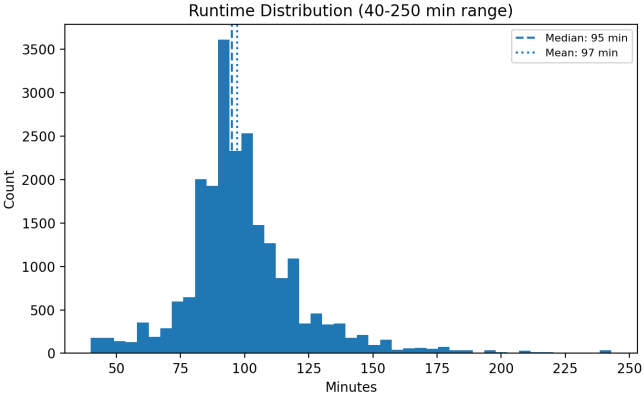
Text Analytics: Review Length and Semantic Signal





Interpretation. Review lengths are long-tailed: the median review is around 173–174 words, but many reviews are much longer. This creates a non-trivial chunking and context-selection problem. The lexical sentiment proxy differs by genre, so semantic predicates may be genre-dependent rather than uniformly distributed.

Structured Analytics: Runtime Distribution



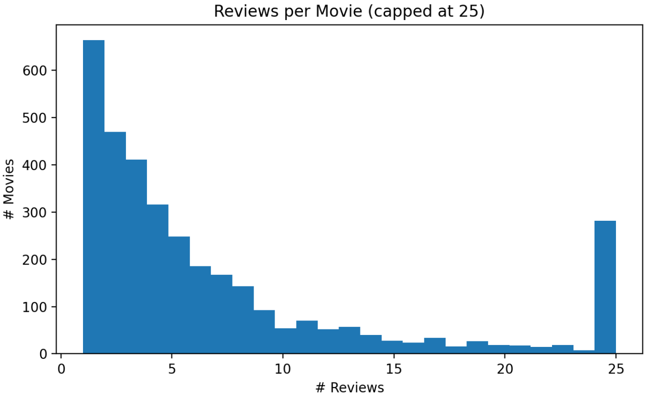
Observations

- median runtime: 95 min
- mean runtime: about 97 min
- most films are concentrated around 80–110 min
- long films are relatively rare but useful for range predicates

Example query slice

```
runtime > 120 AND genre = Drama
```

Entity Granularity: Reviews per Movie



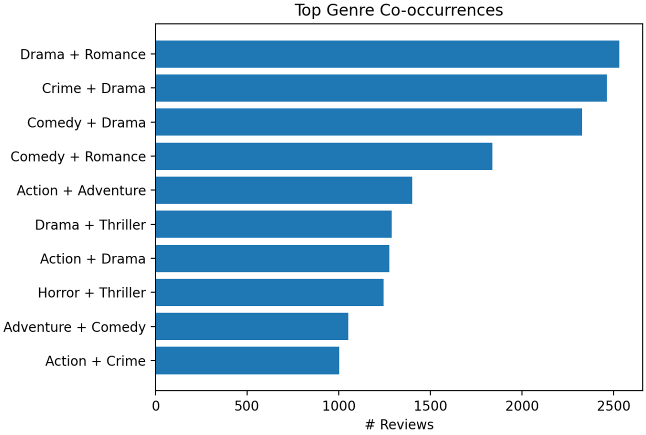
Why it matters

- many movies have only a small number of reviews
- a minority of movies have many reviews
- the dataset has entity-level skew

Implication for systems

For SUQL and Stretto, semantic operators may need either row-level review evaluation or movie-level aggregation before ranking.

Genre Co-occurrences



Observations

- Drama + Romance is the strongest pair
- Crime + Drama and Comedy + Drama are also frequent
- genre labels are multi-valued, not mutually exclusive

Implication

Genre should be treated as a multi-label attribute, similar to an array or ENUM-list column.

Evaluation Implications

The dataset is good, but not balanced. That is useful for realistic system evaluation.

Evaluation slices

- genre: Drama vs long-tail genres
- decade: pre-1980 vs 1980–2011
- review length: short vs long reviews
- entity density: few-review vs many-review movies

Metrics

- final answer precision and recall
- retrieval recall before LLM filtering
- execution cost and latency
- number of LLM calls per query

This allows controlled benchmarking of query execution, not just generic sentiment classification.

Example Query Workloads

The joined table supports questions that cannot be handled well by dense retrieval alone.

Structured + semantic filters

- Which long action movies released after 2000 are described as underrated?
- Which comedy movies have reviews mentioning strong character development?
- Which romance films are described as emotionally powerful?

Why they are hard

- exact constraints should not be approximated by embeddings
- subjective criteria require review interpretation
- ranking should use evidence aggregated over reviews

The useful query plan is usually: structured pruning $\rightarrow$ semantic filtering $\rightarrow$ ranking / aggregation.